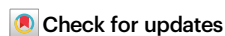


Propagation and preservation of AI-discovered problem-solving strategies in human culture

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Intelligent machines have the potential to uncover problem-solving strategies beyond human discovery. Emerging evidence from competitive gameplay, such as Go and chess, demonstrates that AI systems are evolving from mere tools to sources of cultural innovation adopted by humans. However, the conditions under which intelligent machines transition from tools to drivers of persistent cultural change remain unclear. We identify three key dimensions that modulate machine influence on human problem-solving: the discovered strategies must be non-trivial, learnable, and offer a clear advantage. Using a cultural transmission experiment, we demonstrate that when these conditions are met, machine-discovered strategies can be transmitted, understood, and preserved by human populations, leading to enduring cultural shifts. Conversely, using agent-based simulations, we show how machine influence is constrained in the absence of these conditions. These findings provide a framework for understanding how machines can persistently expand human cognitive skills and underscore the need to consider their broader implications for human cognition and cultural evolution.

Machines powered by Artificial Intelligence (AI) have the potential to discover strategies that are difficult for humans to conceive: their computational capacity exceeds human processing power and speed; their ability to process and share information in serial or parallel with high fidelity allows them to distribute problem-solving efficiently¹. AlphaGo, an AI system that defeated world champion Lee Sedol in the ancient game of Go, discovered superhuman strategies by playing millions of games against a copy of itself, and integrated these experiences into a complex strategy encoded in its neural network². Given the human-defined objective of winning the game, the algorithm used extensive search to identify a successful strategy that has been described as "alien", considering thousands of years of human Go-play³. Recent evidence suggests that humans have significantly

improved their own Go-play as a result of this innovation⁴, while more subtle shifts that appear less "alien" when viewed in historical context are also evident in Go opening moves⁵. In chess, selected strategic concepts extracted from an AI system were applied by grandmasters after observing the AI system only on a few examples⁶. The domain of gameplay remains, however, a rare example of human problem-solving strategies being shaped by machine-discovered strategies. This raises a question: Beyond being mere tools, under which conditions can machines persistently reshape the way we solve problems?

We address this question by integrating insights from the field of Cultural Evolution⁷: culture changes through a process of variation, transmission, and selection^{8,9}. Accordingly, the space of culturally adopted strategies is constrained by human discovery capacity,

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transmission fidelity, and the ability to perceive utility^{10–12}. We propose that machines' capacity to discover novel strategies can expand these boundaries, contingent on three critical factors. *Appropriate Discovery Difficulty*: Strategies must be non-trivial and extend beyond the range of variations typically discovered by humans. *Low Transmission Difficulty*: Humans must be able to learn and effectively disseminate machine-introduced discoveries. *Recognisable Selective Advantage*: Cultural adoption of machine discoveries hinges on their demonstrated success and the extent to which humans attend to them and endorse them. As such, our proposal integrates strands of literature that have remained largely separate: prior research highlighting the transformative power of selective social learning^{13–15}, observational evidence for the general possibility of machine-induced cultural shifts^{2,4,6}, and conjectures about the complementary potential of human limitations and machine capacities¹.

In more detail, for a new problem-solving strategy to become part of the human cultural repertoire, it first needs to be discovered. Human discovery is limited by computational capacities and pervasive cognitive biases^{1,16,17}. For example, human planning is generally confined to a few steps due to the combinatorial explosion of possible actions and outcomes¹⁸. In adapting to this challenge, humans use a range of heuristics¹⁶, such as prioritising immediate over long-term outcomes¹⁹ or favouring familiar options²⁰. In contrast, machines surpass humans in computational capacity, allowing them to rapidly accumulate experiences that would be too costly in terms of time, computation, or risk for humans to access. This difference in resource constraints allows machines to explore much larger hypothesis and action spaces, reducing reliance on strong priors and heuristics that shape human search and enabling the discovery of strategies outside the region typically accessible to human exploration²¹. This may lead machines to discover strategies that are unknown and potentially even counterintuitive to humans, yet have superior performance (see Fig. 1b, vertical axis). Experimental work has yet to investigate how computational heterogeneity and asymmetric exploration capacities shape discovery in human-machine societies, thereby extending prior work which typically considers one agent type^{15,22,23}.

Discovering superior strategies is not sufficient to induce a cultural shift, however²⁴. The superior performance of the machine-discovered strategy must also be easy for humans to recognise as such^{13,25–27}. Furthermore, humans must be able to socially learn the

strategy and transmit it to others⁸. The ability to select superior strategies and to transmit them in combination enables the population to maintain complex strategies that would be beyond individuals' reach^{15,28}. When these mechanisms apply to machine-discovered strategies as well, machines could become sources of new knowledge that shift human culture (see Figure 1b, horizontal axis).

Even though intelligent machines are increasingly used in domains of human discovery, such as education and research^{29–31}, in most cases these AI systems do not convey problem-solving strategies that are entirely alien to human culture. As such, they do not satisfy the above conditions but remain within the confines of existing human discoveries (see Figure 1b dark and medium grey). And even those AI systems that have discovered super-human knowledge often remain, at present, tools in human hands rather than integrated into human cognitive frameworks. AlphaFold provides a remarkable example for this category: while it was inspired by the success of AlphaGo in emulating the intuition of human experts³² and in doing so has achieved solving protein-folding problems at superhuman levels^{33,34}, its discoveries have not (yet) transpired into novel human intuitions, unlike in the case of AlphaGo and AlphaZero. As such, it presents an unlearnable and unpreservable machine discovery (see Fig. 1b, light grey).

Here, we show that machines can persistently reshape human problem-solving culture when they discover strategies that humans typically fail to produce, yet can easily learn and transmit. To test this, we designed a behavioural task in which machines can produce counterintuitive solutions with a recognisable selective advantage. The task emulated a prototypical situation in which human exploration is constrained by the pervasive human tendency to avoid losses (i.e., myopic decision-making). Human participants ($n = 1155$) were paid to solve a series of “reward networks” (Fig. 2c), which we developed as a modified version of an established paradigm^{35,36}. In our task, networks consisted of nodes connected by links that were associated with rewards ranging from -50 to $+400$ points. Participants' goal was to choose 10 moves through the network that would maximise the number of points gained. Our networks were designed such that there was an optimal strategy for solving them: incurring early losses subsequently led to links associated with the maximum reward of $+400$ points, which were not accessible when avoiding early losses. Hence, the optimal strategy for solving our networks contradicts the myopic tendency of humans to avoid losses and prioritise immediate gains,

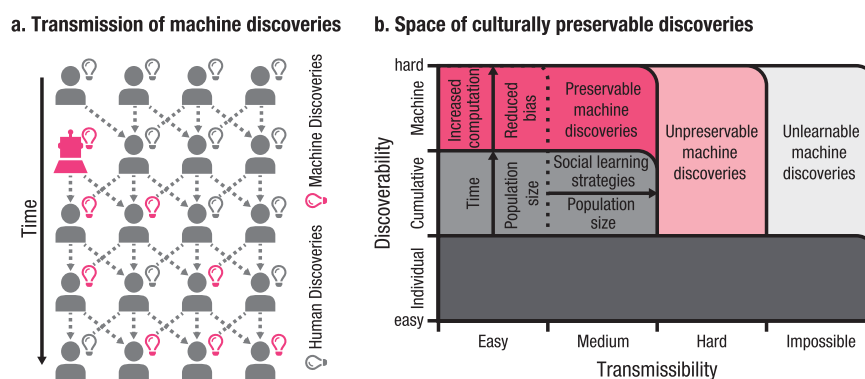


Fig. 1 | Machine-driven cultural shifts. a Transmission of machine discoveries can lead to cultural shifts. The multitude of potential transmission paths enables populations to preserve strategies that are difficult to transmit, fostering resilience against cultural loss. Machines, with their unique capabilities, may discover strategies that are improbable—or difficult to conceive—for humans. If such machine-discovered strategies are successfully transmitted and preserved within human populations, they could drive cultural shifts that outlast the presence of machines. **b Space of culturally preservable strategies.** Human populations exploit strategies that are either easy to learn or sufficiently

transmissible, forming a cultural frontier that has expanded with larger population sizes, time, and the development of social learning mechanisms (dark and medium grey). Machines, by virtue of increased exploratory capacity and reduced inertial bias, may discover strategies that are practically inaccessible to humans. We propose that some of these strategies may nonetheless remain preservable by humans (dark pink). Conversely, other machine-discovered strategies may be neither preservable nor even learnable by humans, potentially fostering a reliance on machines (light pink and light grey).

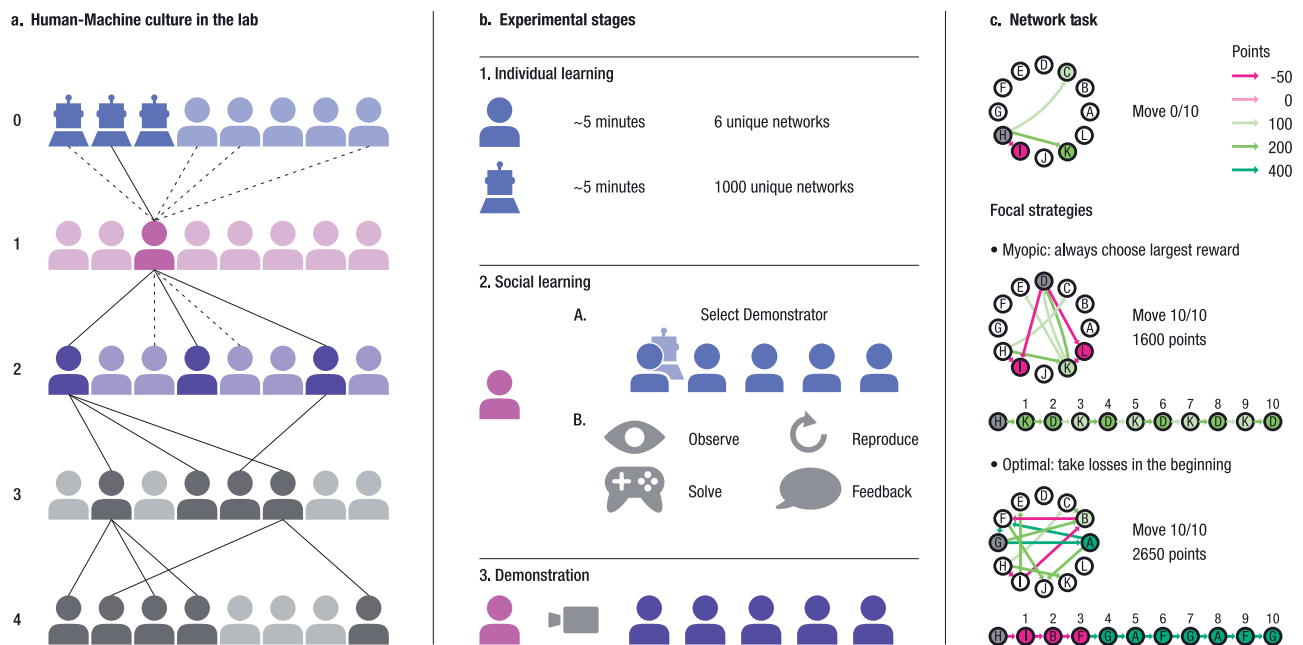


Fig. 2 | Experimental design and network task. a Human-Machine Culture in the Lab. Participants were divided into populations of five Generations (0–4) with eight individuals each, including 15 Human-Machine Populations (three machines in Generation 0) and 15 Human-Only Populations. Lines represent a subset of demonstrator-learner connections (dashed: potential, solid: selected) involving a focal participant (dark magenta) in Generation 1 and their cultural ancestor (dark blue) and descendants (dark violet and grey). **b Experimental Stages.** The 'Individual Learning Phase' lasted five minutes, during which humans in Generation 0 managed to train on six networks, while machines managed to train on 1000 networks. In the 'Social Learning Phase', Generations 1–4 were informed about the

performance of 5 randomly drawn players from the previous generation (demonstrator candidates), from which they selected their preferred demonstrator. They then observed and replicated the demonstrator's solutions before solving the same network themselves and receiving feedback. In the 'Demonstration Phase,' individuals recorded their solutions for the next generation. **c Network Task.** Participants collected points on a network, with new potential connections appearing sequentially. We illustrate the two focal strategies in our task: maximising immediate gains (myopic), and incurring initial costs for greater future rewards (optimal).

even when accepting short-term losses could be beneficial in the long run. In line with prior literature, we expected that this myopic tendency would make the discovery of an optimal strategy in our task difficult for human players¹¹. Meanwhile, we expected that a few minutes of training on our task would be sufficient for a deep-Q-learning algorithm (the "machine player") to successfully discover the optimal strategy through unbiased exploration at scale. To compare the performance of human players on their own against the performance of human players exposed to machine players, we randomly allocated participants into populations that involved either only eight humans (15 "Human-Only Populations") or five humans and three machine players in their initial generation (15 "Human-Machine Populations"). Each population consisted of five generations (numbered 0–4; Fig. 2a), allowing us to observe differences between the two kinds of populations over time. After a series of six individual attempts at solving networks, participants in Generation 0 solved four networks during the "Demonstration Phase", knowing that their moves would be recorded for future generations to learn from. After two individual trials to familiarise themselves with the tasks, participants from Generations 1–4 could choose a "demonstrator" from the previous generation to learn from, based on the demonstrators' average performance. They did not know the demonstrators' identity as a human or machine, but were informed that machine players were a possibility. They then observed animations of the demonstrator's moves, replicated them, and subsequently solved the same networks before receiving final feedback, with this procedure repeated across four networks. Only after this Social Learning Phase did participants from Generations 1–4 demonstrate their moves for the next generation. All networks solved by the participants throughout the experiment were unique, requiring them to discover or learn a generalisable strategy rather than optimising a specific solution.

In a second exploratory step, we sought to follow up on our experimental results with an agent-based simulation. Removing the feasibility constraints that apply to large-scale experiments with human participants, the simulation allowed us to systematically vary discovery and transmission difficulty, and to manipulate the extent to which social learning is selective¹⁴. In parallel to the experimental set-up, we organised agents into five generations, each consisting of eight agents. Agents could discover strategies through exploration or adopt strategies from the previous generation. Abstracting away from the task we used in the experiment, agents explored a simplified strategy space comprising the strategies: *myopic* and *optimal*. For "machine" agents, we independently manipulated exploration capacity and bias to investigate their impact on the strategies adopted within hybrid populations across subsequent generations.

Results

Humans preserve the optimal strategy discovered by machines

In our reward network task, achieving more than 2000 points was only possible by following the optimal strategy of incurring early losses to reach later links associated with the maximum reward of +400 points per move. Generally, human players did not find this optimal strategy on their own: Among the human populations, only one out of 600 participants *discovered* the optimal strategy over the course of the experiment, which was subsequently learned by four other human players. While the deep Q-learning agent, through extensive exploration over 20,000 epochs across 1,000 networks, discovered and reliably reproduced the optimal strategy within less than five minutes (Supplementary Fig. S7), human participants, by contrast, exhibited a systematic avoidance of loss-incurring

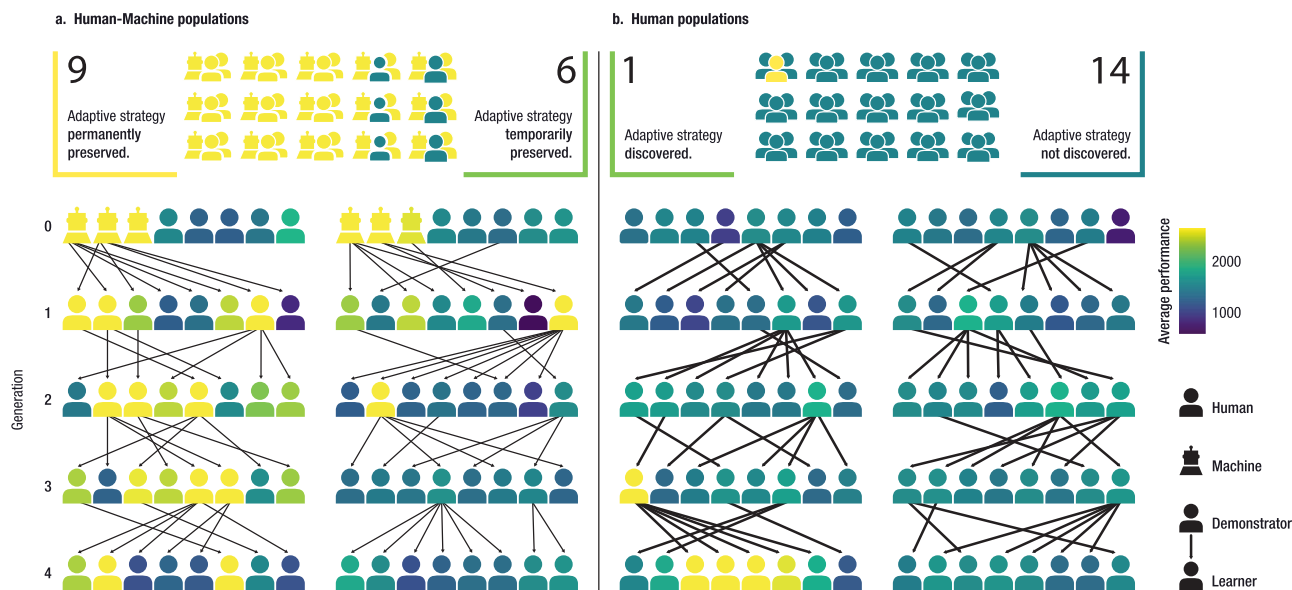


Fig. 3 | Four prototypical populations. We categorised populations based on the average scores of their top performers across generations. Scores above 2000 points indicate adoption of the optimal strategy. Among 15 Human-Machine Populations (panel a), 9 consistently exceeded this threshold, while the remaining 6 surpassed it temporarily in at least one generation following the introduction of

machines. In contrast, only 1 of 15 Human-Only Populations (panel b) temporarily achieved scores indicative of the optimal strategy, while all the other Human-Only Populations did not. We illustrate one prototypical population for each case, with colours reflecting performance scores. Arrows represent demonstrator-learner relationships.

trajectories: paths involving three losses were under-sampled relative to a matched random baseline (LOR = -1.22 , $CI_{95\%} = [-1.38, -1.06]$), reducing the likelihood of discovering the optimal strategy (see also Supplementary Text). Despite the rarity of individual human discovery, the optimal strategy readily spread from machines to humans in the Human-Machine Populations and was preserved beyond the presence of the machines across multiple generations: In 9 out of 15 Human-Machine Populations, human players learned this strategy in each of their generations. Correspondingly, we classified these populations as *permanently preserving* the machine discovery. In the remaining 6 populations, at least one human player in the generation immediately following the machine achieved more than 2000 points, but this discovery was only *temporarily preserved* as it did not transmit to later generations. Figure 3 illustrates prototypical cases for each of these qualitative categories. To summarise, the majority of Human-Machine Populations preserved the optimal strategy until the final generation in our experiment, while 14 out of 15 human populations did *not* discover the optimal strategy at all.

Preservation of the optimal strategy was supported by a bias toward selecting the demonstrator with the highest performance. Overall, 86.7% ($CI_{95\%} = [84.5, 88.7]$) of all participants chose the highest-scoring demonstrator. Consequently, regarding the cultural lineages emerging from demonstrator-learner relationships within the Human-Machine Populations (see Fig. 3), we found that only machines had long-lasting impact on the cultural dynamics operating across generations: In the first socially learning generation of the Human-Machine Populations, most players (90%) selected a machine as their demonstrator, and by the last generation, every player was a descendant of a machine player. In contrast, none of the human players in Generation 0 of the Human-Machine Populations initiated a cultural lineage that persisted until the last generation.

Humans benefit from the machine discovery

Panel A in Fig. 4 shows the average reward gained in the task by generation and population. Descriptively, while the two types of populations started out with similar descriptive averages in Generation 0, in

Generations 1–4 Human-Machine Populations consistently earned higher rewards. Our preregistered linear mixed-effects model on these latter generations supported that this difference was significant ($\beta = -309.2$, $SE = 57.0$, $CI_{95\%} = [-420.7, -196.6]$, $P < 0.001$). This model corresponds to model 1a for reward in the preregistration and includes an interaction between condition and generation (like model 2b). Further models confirmed that Human-Machine Populations performed better than Human-Only Populations both in Generation 1, which could learn directly from machine players (reward model 2a: $\beta = -404.6$, $SE = 59.4$, $CI_{95\%} = [-513.5, -293.0]$, $P < 0.001$), and in Generation 4, where influence from machines was remote by four iterations (reward model 1b: $\beta = -229.1$, $SE = 82.4$, $CI_{95\%} = [-407.4, -72.3]$, $P = 0.010$).

Humans are behaviourally congruent with the machines

Panel B in Fig. 4 shows the proportion of human moves congruent with machine-generated moves, summarised by generation and population. Qualitatively, the results are similar to the results for rewards: no statistically significant difference between conditions in Generation 0, but Human-Machine Populations are consistently more behaviourally congruent in Generations 1–4. A logistic mixed-effects model on these generations shows that Human-Machine Populations were more congruent than human populations (congruency model 1a: $\beta = -0.87$, $SE = 0.14$, $CI_{95\%} = [-1.12, -0.61]$, $P < 0.001$). Again, this was also the case within Generation 1, which learned from machine players (congruency model 2a: $\beta = -1.13$, $SE = 0.13$, $CI_{95\%} = [-1.41, -0.88]$, $P < 0.001$), and Generation 4 (congruency model 1b: $\beta = -0.64$, $SE = 0.20$, $CI_{95\%} = [-1.02, -0.25]$, $P < 0.001$). Notably, the individual who independently discovered the optimal strategy, as well as subsequent participants, likewise exhibited elevated congruency with machine behaviour (see Supplementary Fig. S9).

Humans internalise the machine discovery

During the Demonstration Phase of our experiment, players were asked to provide a written strategy of how they approached the task. In contrast to the demonstration itself, these written strategies were not

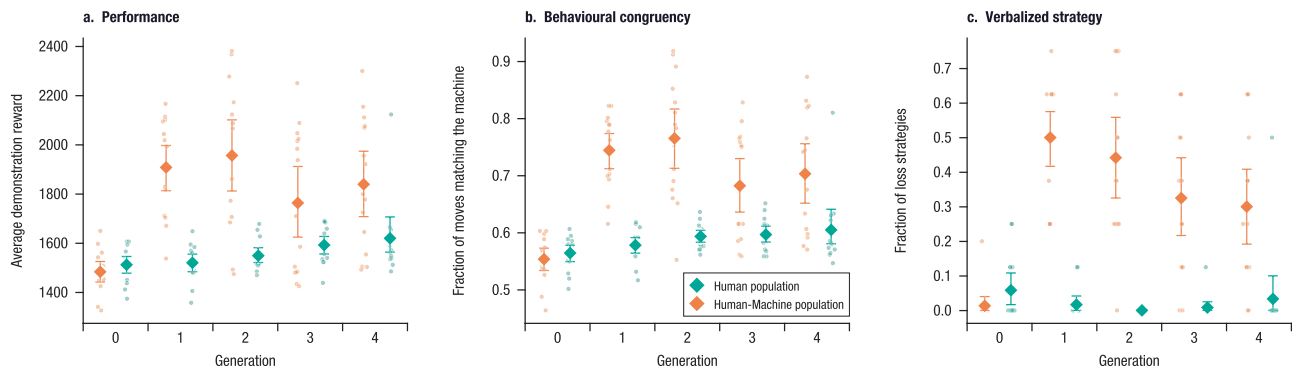


Fig. 4 | Evolution of task performance, behavioural congruency, and strategy description of human participants. Individual populations are represented as dots, with the grand average for each condition indicated by a diamond symbol. Human-Only Populations are indicated in green, Human-Machine Populations in orange. Performance (Panel **a**) refers to the average number of points earned during the Demonstration Phase. Behavioural Congruency (Panel **b**) indicates the proportion of human player moves that are congruent with the majority behaviour of the three trained machines. Verbalised Strategy (Panel **c**) shows the fraction of

human players who articulated a strategy referring to a deliberate acceptance of losses after the Social Learning Phase. Data are presented as mean values. Error bars indicate 95% bootstrap confidence intervals (CI) around the mean, computed over population-level aggregates. $n = 15$ populations per condition, each comprising 8 participants per generation. Each dot represents one population. The Human-Only condition serves as the control group. Participant-level means are reported in Supplementary Fig. S11.

transmitted to the next generation, and players were aware of this. Panel C in Fig. 4 shows the proportion of human players mentioning the optimal strategy (i.e., the "loss strategy") in their written statements, as coded independently by three human annotators. In the Human-Only Populations, we observe practically no mention of the strategy, with proportions close to 0 throughout all generations; the exception to this is the single population that discovered the optimal strategy in Generation 3, resulting in 50% of players referring to it in Generation 4. In contrast, participants in the Human-Machine Populations consistently mentioned the optimal strategy explicitly, with proportions of about 30% to 50% per generation. A logistic mixed-effects model on Generations 1 to 4 shows that this difference is significant (model 2b: $\beta = -4.18$, $SE = 0.62$, $CI_{95\%} = [-5.49, -3.04]$, $P < 0.001$).

Simulation corroborates conditions for cultural shift

We devised an agent-based simulation to systematically examine how variation in discovery and transmission difficulty shapes the cultural impact of machines operating at superhuman capacity. Agents attempted one of two strategies, myopic or optimal, over a finite number of trials K , and discovered the corresponding strategies through a binomial process with success rate d . A bias parameter q controlled the likelihood of attempting the optimal rather than the myopic strategy. We simulated 20 generations of 20 agents each, in which agents could also acquire strategies through social learning from the previous generation with success rate t .

We first confirmed that, in homogeneous populations and under high social learning success rates $\lambda = 0.9$, discovery of strategies of higher difficulty is constrained by bias q , limited exploratory capacity K , and the number of generations elapsed, G (see Supplementary Fig. S12). We then focused on scenarios with heterogeneous exploration capacity and bias, in which the majority of agents (i.e., human agents) are limited in their exploration capacity $k_m = 10$ and exhibit a bias $q_m = 0.01$ toward myopic solutions. We systematically manipulated the exploratory capacity and bias of five machine agents added to the first generation and examined the resulting adoption of optimal strategy by human agents in the final generation (Fig. 5). Increasing the exploratory capacity $K_m \in [1000, 100000]$ of the machine agents and neutralising their bias toward myopic strategies $q_m = 0.5$ both increased the likelihood of discovering the optimal strategy. Consequently, introducing such machine agents enabled populations to

adopt optimal strategies even under high discovery difficulty and thus expanded the horizon of obtainable strategies within a population.

The simulations further show that when optimal strategies are difficult to transmit, they fail to spread and persist across generations, thereby constraining machine-induced cultural shifts. This effect is amplified when agents are less able to identify and select successful models for social learning, either because they are insufficiently sensitive to performance differences (higher τ) or because realised rewards are highly noisy (higher ϵ). Under these conditions, optimal strategies neither diffuse nor persist within populations (see Supplementary Figs. S13, S14).

Discussion

Our work demonstrates that AI discoveries can persistently reshape human problem-solving strategies when they are calibrated in difficulty, transmissibility, and selective advantage. In our experiment, exploration at a capacity several orders of magnitude greater allowed machine players to discover a superior strategy that offset human cognitive biases toward myopic and suboptimal solutions and, as such, was effectively beyond human reach within the constraints imposed by the experiment. This does not mean that the optimal strategy was impossible to discover for human participants at all, but limitations in time and cognitive biases¹ made it exceedingly rare. Yet, the machine-discovered strategies were still learnable for human players, who did not merely replicate them but also applied them to new variations of the task. Further signifying the integration of the counterintuitive problem-solving strategy into their cognitive framework, many human players explicitly articulated the strategy in writing. The machine discovery's selective advantage translated into human players favouring machine over human ancestors, producing persistent cultural lineages where every human player in the last generation was ultimately a descendant of a machine player in the first generation. Consequently, the optimal strategy was adopted by the majority of populations with machine ancestors, whereas human populations predominantly employed the myopic strategy, indicating the capacity of machines to act as drivers of cultural change.

Human culture supports the cumulative accumulation of knowledge and skills across generations, enabling individuals to benefit from collective experience that far exceeds what any single person could acquire in isolation^{9,10}. Laboratory studies explicitly compressing generational timescales into hours or minutes show that social transmission can overcome time constraints and the costs of individual

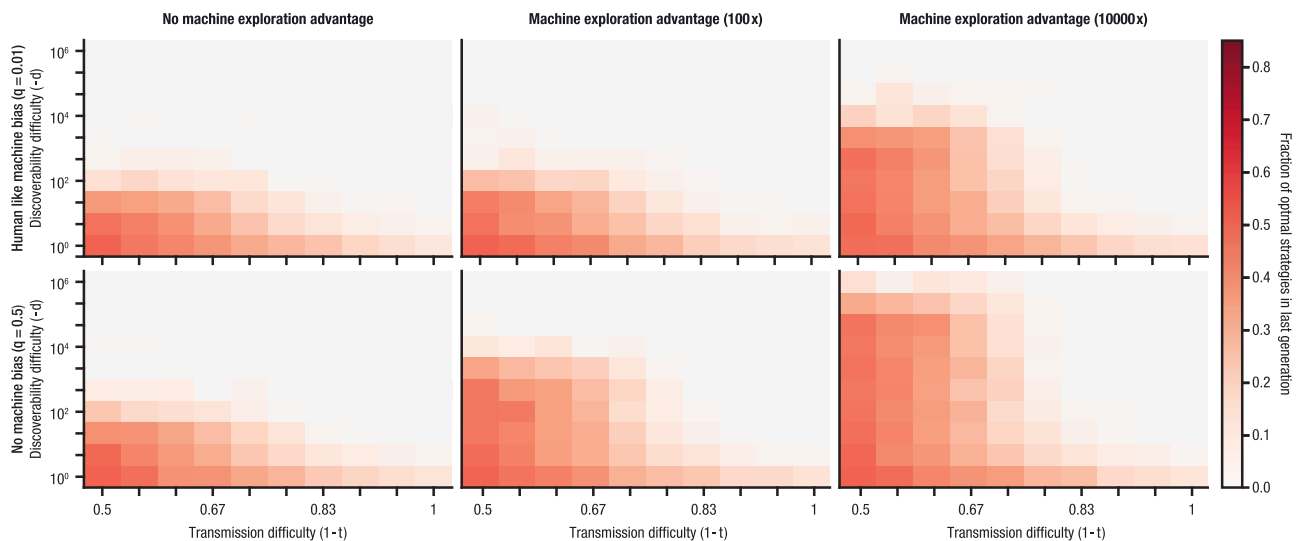


Fig. 5 | Agent-based simulation highlights conditions for cultural shifts. We developed an agent-based simulation replicating our multigenerational social learning experiment (20 generations of 20 agents), systematically varying the difficulty of discovering and transmitting the optimal strategy. Five “machine” agents were introduced in the early generations, and we manipulated their exploratory capacity and bias toward myopic solutions relative to “human” agents. The heatmap shows the proportion of agents in the final generation (in the absence of

machines) that adopt the optimal strategy. The upper-left panel is equivalent to a scenario without machines. Increased exploration and reduced bias in machine agents enable the discovery of strategies beyond human reach, which are subsequently adopted and maintained by human agents. Colour scale represents the fraction of optimal strategies in the last generation, ranging from 0 (light) to 0.8 (dark).

exploration^{37,38}. Building on this literature, the present work suggests that machines may function as novel sources of cumulative cultural input. By learning from machines, humans may adopt and transmit hard-to-conceive optimal strategies, thereby expanding their cognitive repertoire without incurring the cognitive costs and time typically associated with discovering such strategies. Our results apply particularly to scenarios in which human exploration is constrained, while machine exploration remains scalable. For instance, chemical recipes can be explored at high throughput and scale through automation³⁹. In our experiment, the state space was small enough to permit brute-force exploration, and the primary role of Q-learning was to extract a generalisable, model-free strategy from this experience. In many real-world domains, however, solution spaces are intractable, and exploration is resource-intensive. Future research should examine whether the extensive world knowledge embedded in large generative models might similarly facilitate previously inaccessible discovery processes when both humans and machines operate under comparable data constraints.

Despite most populations maintaining the algorithmically discovered optimal strategy, the results show that even selectively advantageous strategies remain vulnerable to stochastic loss under finite populations and transmission noise. Prior work in cultural evolution indicates that large effective population sizes can buffer against the loss of complex discoveries^{22,40}, suggesting that machine-induced cultural shifts may be amplified when large groups interact with machine-generated strategies, and transmission is the main constraint. For instance, the widespread adoption of generative AI in educational, professional, and creative contexts exposes large populations to machine-mediated strategies. Larger groups increase the likelihood that such strategies are learned, as skilled individuals transmit them to others through repeated demonstrations²², facilitating the accumulation of a critical mass necessary for tipping points in cultural adoption⁴¹. By contrast, in smaller populations, machines’ advantages in discovery can be particularly pronounced, given the limited collective exploration of such human groups. Likewise, emerging evidence suggests that machines’ ability to rapidly identify adaptive solutions within a population can enhance collective efficacy⁴².

Yet another factor that could broaden the scope of machine-induced cultural shifts is pedagogical behaviour. Teaching is one of the primary mechanisms through which humans transmit knowledge and skills^{8,43}, and experiments have shown that the value of teaching increases with the complexity of strategies⁴⁴. Notably, discovery and transmission difficulty have also been identified as key determinants in the evolution of teaching⁴⁵, underscoring that machine-induced cultural shifts may occur within a relatively narrow range of conditions. This line of work does, however, also point towards another possible pathway that could enhance the efficacy of machine-induced cultural shift: more explainable AI systems could provide both teachers and learners with insights that facilitate the adoption of machine-discovered strategies, and enhance their scope through more effective human-machine interactions⁴⁶. In addition, recent work in chess suggests that testing for AI-to-AI teachability could be a promising way to identify strategic machine concepts that can be effectively transferred to humans⁶. At the same time, pedagogical features in AI systems may not always be necessary or desirable, particularly when humans use these systems primarily as tools. As AI systems are increasingly trained on one another’s outputs, they may transmit adaptive strategies in the form of cultural artifacts that are opaque to humans, potentially leading to a bifurcation of human and machine cultures⁷.

A major question arising from these results concerns the scope of machine-induced cultural shifts. In pilot experiments, we found a key challenge in achieving the right balance between discovery difficulty and transmissibility (see Supplementary Text): At one extreme, simple strategies were readily discovered by humans without machine assistance, limiting their capacity to induce meaningful shifts. At the other extreme, strategies requiring lengthy and precise sequences of actions were too complex for human participants to learn and pass on. It is remarkable that the optimal strategy in our final design was discovered by none of the 195 human participants in the first generation (in deviation from previous work, such as ref. 15, where the discovery rate in the first generation was still at 13%). Nonetheless, in our setting, this optimal strategy was still learnable and preservable once it was discovered. As a controlled online experiment, our design necessarily

reflects stylised conditions, with individual learning constrained to brief exploration periods that may differ from real-world settings. However, analyses of participants in generation 0 with greater learning opportunities provide no statistically significant evidence that extended exploration budgets attenuate loss-related biases (see Supplementary Text).

Theoretical frameworks about cumulative cultural evolution posit that culture evolves through the iterative refinement of adaptive solutions and strategies⁴⁷, implying that certain outcomes may remain undiscoverable without intermediate adaptive steps. The results presented in this work suggest that the scale of exploration alone can enable machines to discover strategies that humans fail to uncover, but are able to learn. They also indicate an interplay between exploration scale and bias, or cultural entrenchment, that constrains human exploration. Building on this insight, future studies could explore how the enhanced exploratory depth and breadth enabled by machines might help overcome structural limitations, such as entrenched path dependencies and biased expectations shaped by a historically conditioned epistemic horizon^{31,48}. Supporting this conjecture, recent work has highlighted machines' potential to identify novel strategies for fostering cooperation that humans have proven unlikely to consider⁴⁹. Even when it comes to changing notoriously entrenched conspiracy beliefs, intelligent machines (here, LLMs) are emerging as potent agents of persistent learning among humans⁵⁰. But these contexts also point to the potential challenge that machine-discovered strategies might sometimes be undesirable from a human perspective and not be explored for good reasons, such as ethical concerns or social acceptability. Relatedly, human adoption and preservation of machine-discovered strategies may raise ethical concerns about human autonomy and creativity as much as it might bear epistemic risks that could undermine human knowledge⁵¹. In other contexts, AI aversion or other (oftentimes variable) perceptions of machine-derived knowledge^{36,52,53} might influence adoption rates. However, as it becomes increasingly difficult to dissociate human- from machine-derived cultural artifacts⁵⁴, it is all the more crucial to understand how machines might expand the frontiers of human thought and enable the emergence of culturally significant, hard-to-conceive strategies.

Methods

This study was approved by the Ethics Committee of the Max Planck Institute for Human Development (protocol A 2022-24). All participants provided informed consent prior to participation.

Participant Recruitment, Exclusion and Compensation

Participants ($n = 1155$) were recruited through the online platform Prolific⁵⁵, while screening for individuals residing in the United States and the United Kingdom, being proficient in English, and maintaining an approval rating of at least 95%. Upon successful completion of the study, participants received a fixed payment of €2.25, complemented by performance-dependent bonus compensation payments averaging €0.70. The median time for completing the study was 13:45 minutes. The average hourly compensation rate was €9.82.

Participants ranged in age from 18 to 92 years ($M = 39.67$, $SD = 12.28$). Of those with available demographic data ($n = 1272$), 56.4% identified as male, 43.2% as female, and 0.3% preferred not to say. The majority resided in the United Kingdom (72.9%), versus the United States (27.1%). Sex and gender were not analysed as variables in this study as they were not relevant to the research question.

As preregistered, participants were excluded based on two criteria: (1) Participants who took more than 25 minutes, deemed unreasonably long; (2) Participants with missing moves in any of their demonstration trials, as subsequent generations could not replicate incomplete solutions. These criteria also implicitly served as an attention check. Of the 1303 participants who completed the

experiment, 148 were excluded under these conditions. Their spots were then filled by newly recruited participants. No other exclusions were made. The investigators were not blinded to allocation during outcome assessment, as the analysis followed a preregistered plan.

Participants who passed the exclusion criteria were eligible for bonus compensation. Participants' compensation depended in part on their own performance in the Demonstration Phase, and in part on the performance of people that learned from them. To ensure a fair chance of gaining a bonus payment independently of the condition a participant was sampled in, we ranked these scores separately per generation and condition. The final bonus was then computed as the percentile score rank multiplied by €1.40.

Preregistration

The study was preregistered on asPredicted.org on 15 February 2024 (<https://aspredicted.org/d765-77fm.pdf>), including a detailed description of predictions, measures, conditions, analyses, exclusion criteria and sample size. No deviations from the preregistered analysis plan were made. In particular, we preregistered the following predictions:

1. Main predictions

- General benefit of AI.* Overall, participants from generation 1 onwards in AI trees will outperform participants in human trees.
- Long-lasting benefit of AI.* The subset of participants in the final generation of AI trees will outperform the final generation of the human trees.

2. Secondary predictions

- Transmission from AI to humans.* The subset of participants in generation 1 of the AI trees will outperform generation 1 in the human trees.
- Explicit recognition of counterintuitive strategy.* Overall, participants from generation 1 onwards in the AI trees will describe the counterintuitive strategy more frequently in their written strategies than those in the human trees.

Network task

We adapted the Reward Network task^{35,36} consisting of a network g with 12 nodes and 30 edges, where each edge is associated with a reward $r \in [-50, 0, 100, 200, 400]$ (see Fig. 2 panel c). Hidden to participants, each node was associated with a level $l \in [0, 1, 2, 3]$. Constraints in the network generation ensured that any path connecting a node of level 0 to a node of level 3 included at least three edges with negative rewards. The largest reward of 400 points was accessible only from level 3. These two constraints required participants to accept at least three losses to access the highest rewards. Transitions between nodes of level 0 contained the second-highest reward of 200 points, making myopic behaviour a reasonable but generally suboptimal strategy.

Participants began the game at a predetermined starting node (e.g., node B in figure 2 panel c). Initially, only the outgoing edges from this node were revealed. As participants traversed the network, they sequentially uncovered it, and the outgoing edges of each visited node were displayed. Each participant would be playing on, in total, 10 different networks drawn randomly from the pool of experimental networks. The assignment of networks to individual participant trials ensured that no participant experienced the same network twice.

We generated three sets of networks (training, validation, and experiment) with 1000 networks each. The first two sets were used only during the training of the neural policy, while the last set was used exclusively during the experiment. After generation, we computed scores for a myopic rule-based strategy and a loss-seeking rule-based strategy. The myopic strategy always selects the move with the immediate highest reward, while the loss-seeking strategy prioritises edges with a -50 reward over all others. We excluded networks from all three sets where the myopic strategy outperformed the loss-seeking

strategy, which accounted for approximately 9% of all generated networks.

Populations and Social Learning Opportunities

The experiment consisted of 30 populations, each with 5 generations and 8 players per generation (see Fig. 2a). Half of the populations were assigned to the "Human-Machine" condition, and the other half to the "Human-Only" condition. In the "Human-Machine" condition, machines took three player positions in the first generation. Starting from generation 1, each participant could choose one of five possible demonstrators drawn from the eight positions in the previous generation. Participants were dynamically assigned to available positions in the populations, prioritising the completion of existing populations before adding participants to new ones. Positions for later generations became available only after all potential demonstrators had completed the experiment.

Experimental flow

Panel B in Figure 2 depicts the main phases of the experiment. A static version mimicking the experience for participants in later generations is available here: <https://center-for-humans-and-machines.github.io/reward-network-iii>.

Below, we summarise the main phases of our experiment:

- 1. Introduction Phase:** Participants learned about the study's purpose, procedures, data handling, estimated time, and financial compensation. They then completed a tutorial guiding them through the task and interface.
- 2. Individual Learning Phase:** Participants solved the task individually. Generation 0 worked through 6 networks, while later generations worked through only 2 networks (given they encountered four more in the Social Learning Phase, see below) to familiarise themselves with the task. See also Fig. S3.
- 3. Social Learning Phase (only from generation 1):** Participants selected a demonstrator (see Supplementary Fig. S4). They could base their decision on demonstrators' average scores from the Demonstration Phase (they were also informed about their own average score as a reference point). They then went through four networks by observing a replay of the demonstrator's solution (see Supplementary Fig. S5), repeating that solution (see Supplementary Fig. S6), and trying to solve the same network on their own. Replay delays between moves were equalised to eliminate timing differences between human and machine-generated demonstrations. Last, they received feedback comparing their solution with the demonstration.
- 4. Demonstration Phase:** Participants independently solved four networks. The solutions entered were used for the Social Learning Phase of the next generation for those selected as a demonstrator.

Participants also documented their personal strategy in written form, before and after the Social Learning Phase. These strategies were collected solely for later analysis and were not presented to subsequent generations.

Neural policy and its training

We used a deep Q-learning-based reinforcement learning approach to train a neural policy for navigating and solving reward network tasks, as outlined by ref. 56. The neural policy featured a gated recurrent unit (GRU)⁵⁷, flanked by two linear layers with ReLU activation, each containing 15 hidden units. The input $Q(o_t)$ consisted of a one-hot-encoded reward of each target node in a vector of shape $\text{Nodes} \times \text{Rewards}$, while the output represented the Q-value of each target node, with unreachable nodes set to negative infinity. Thus, similar to the initial experience of human participants, the algorithm could only process the immediate outgoing edges from the current node without engaging in any explicit planning.

The neural network approximated Q-values $Q(o, a; \theta)$ where θ represented the network parameters. These parameters were optimised by minimising the loss function:

$$L(\theta) = \mathbb{E} \left[(y_t - Q(o, a; \theta))^2 \right], \quad (1)$$

with the target value y_t defined by the Bellman equation:

$$y_t = r + \gamma \max_{a'} Q(o', a'; \theta^-), \quad (2)$$

and $\gamma = 0.99$ as the discount rate. The target policy weights θ^- were only updated every 200 steps to stabilise training.

An epsilon-greedy strategy guided training with ϵ starting at 1 and reducing according to $\epsilon_e = 0.99^{[e/1,000]}$, reaching a minimum of 0.01. Trajectories were stored in a replay buffer of 500 episodes, and a batch of 16 episodes was sampled for updating the policy. The policy was optimised using the Adam optimiser with an initial learning rate of 1×10^{-3} , adjusted downwards by a factor of 0.8 every 2000 episodes. The agent underwent 20,000 episodes of training, with evaluations every 100th episode using 1000 test networks from a hold-out set. Training took approximately 5 minutes on an RTX 5000 GPU. The neural policy was trained using Python 3.10 and PyTorch 2.1.1. Complete package versions are listed in the code repository. We depict the average performance on a hold-out test set in Supplementary Fig. S7, illustrating that the policies initially discovered the myopic strategy before eventually settling on the optimal strategy.

Measures

Task performance. We measured task performance by calculating the total score for each trial, which involved summing the rewards from the 10 moves. The maximum achievable reward was 2650, which corresponded to taking three losses of 50 points each and then gaining 400 points seven times. In contrast, the maximum score attainable with a myopic strategy was 2000, yielded by ten gains of 200 points each. The exact scores achievable with these two strategies varied between networks.

Behavioural congruency. We assessed human-machine behavioural congruency by comparing the actions taken each human l , denoted as $a_{l,t}^H$, with those predicted by an ensemble of three machines. For each machine $k \in \{1, 2, 3\}$, the predicted action was defined as

$$a_{l,t}^{M_k} = \arg \max_a Q_k(a_{l,0}^H, \dots, a_{l,t-1}^H), \quad (3)$$

based on the trajectory of moves $a_{l,t}^H$ selected by the human participant. The ensemble prediction was then determined by majority vote,

$$a_{l,t}^M = \text{mode}(a_{l,t}^{M_1}, a_{l,t}^{M_2}, a_{l,t}^{M_3}). \quad (4)$$

A match $a_{l,t}^H = a_{l,t}^M$, where the ensemble's action corresponded with the human's action, was encoded as '1'; a mismatch as '0'. This method required processing the entire trajectory to ensure that the recurrent neural units of the neural policies could utilise accumulated historical information to influence their decisions.

Written strategies. To measure the explicit recognition of the loss strategy, we manually coded all 2310 strategy descriptions (covering both time steps for every participant) with the aid of three trained human annotators. The annotators familiarised themselves with the task and independently coded the written strategies using a binary variable to indicate the presence of the loss strategy. Before coding the full experimental data, the annotators calibrated their ratings by coding a small independent dataset ($n = 76$) from two previous pilot studies. After individual coding, interrater agreement was high (Light's

$\kappa = 0.83$). They then discussed any discrepancies before proceeding to code the full dataset. Annotators were blind to participant conditions during both calibration and full data coding, and data was randomly shuffled prior to coding to prevent any bias linked to participant order. For the full dataset, interrater agreement was excellent (Light's $\kappa = 0.92$).

Statistical procedure

All analyses were performed using the *lme4* package (version 1.1-35.1;⁵⁸) in R (version 4.3.2; R Core Team), and the structure and procedure of these analyses were part of our preregistration. We first subset the data to only include demonstration trials of participants. Next, we aggregated the data to the appropriate level for each measure of interest: trial-level for reward, move-level for behavioural congruency, and participant-level for the written strategies.

In all models, the predictor of interest was a fixed effect of *condition*, assessed by computing the bootstrapped confidence intervals around the estimate at 95% confidence level. All confidence intervals are two-sided at the 95% level. For the models predicting the measure across generations 1–4, we included random intercepts for *participant* and *population*, along with a random slope for *generation* within *population*. For models predicting measures within individual generations, we incorporated the random intercepts for *participant* and *population*. Reward outcomes were modelled using a linear link function, whereas behavioural congruency and strategies were modelled using a logistic link function. As preregistered, we centred and scaled the variable *generation* to enhance model convergence and to make the main effect of *condition* more interpretable. Residual diagnostics for the linear mixed-effects models confirmed approximate normality and homoscedasticity of residuals.

Power analysis

We determined the goal of 15 trees per condition by simulating data via an agent-based model resembling a variant of the model described in the next section. In this model, rates of learning success and individual discovery rates were calibrated based on pilot data. We created synthetic data for 100 independent runs of the full experimental design in this way. In a power analysis on this simulated data, we resampled runs with 10 or 15 populations, each 1000 times. We then checked the distribution of parameters for the main preregistered models 1a and 1b, and based our final decision on 1b (since it has the smaller subsample of the two main predictions). We found our predicted effect, as assessed by a t-value greater than 2 for “condition”, in 957 (96%) of the samples involving 15 populations, suggesting sufficient power to assess all predicted effects.

Agent-based simulation

We developed a parsimonious agent-based model to capture key mechanisms of multi-generational cultural transmission. Each generation consists of sequential individual learning, reward-based teacher selection, social transmission, strategy integration, demonstration, and reward updating.

Simulations comprised $M = 50$ replications, $G = 20$ generations, and $N_{\text{gen}} = 20$ agents per generation. Depending on the condition, $N_m = 5$ agents were machines with systematically manipulated exploration capacities K_m and optimality biases q_m .

During individual learning, human agents performed k_h trials, allocating exploration between optimal and myopic strategies according to an exploration bias q_h . Optimal and myopic strategies were discovered with cumulative probabilities determined by per-trial rates d_{opt} (manipulated) and $d_{\text{myo}} = 0.5$, reflecting the greater accessibility of the myopic solution. Agents adopted an optimal, myopic, or null strategy, with priority given to the optimal strategy when both were discovered. Successful agents produced $K_{\text{demo}} = 10$

demonstrations of the adopted strategy. Rewards were computed as

$$R = n_{\text{opt}}R_{\text{opt}} + n_{\text{myo}}R_{\text{myo}} + \epsilon_{\text{eff}}, \quad \epsilon_{\text{eff}} \sim \mathcal{N}(0, \epsilon^2 K_{\text{demo}}), \quad (5)$$

with $R_{\text{opt}} = 1$, $R_{\text{myo}} = 0.5$, and $\epsilon = 0.1$.

The initial generation relied exclusively on individual learning. For generations $g > 0$, learners sampled candidate teachers from the previous generation and selected among them via softmax weighting over rewards with temperature τ . Social transmission occurred with probability λ . Individual and social learning outcomes were integrated such that optimal strategies dominated whenever acquired. Provided that evolutionary selection operates through teacher choice, we adopt this simple prioritisation of demonstrated behaviour, assuming that agents aim to maximise individual performance during demonstration.

We examined homogeneous and mixed populations across ranges of exploration bias, discoverability ($-\log_{10}d_{\text{opt}}$), and learnability ($1 - \lambda$), and systematically varied machine capacity and bias. Additional analyses varied selection strength τ , reward noise ϵ , machine placement, population size, and machine prevalence to assess robustness. Full implementation details and extended analyses are provided in the Supplementary Information.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All experimental data are deposited on Zenodo (<https://doi.org/10.5281/zenodo.18547122>)⁵⁹. The deposit contains both raw experimental data, as downloaded from the experiment interface, and processed data with derived variables used in the analyses. All data are de-identified and provided at the individual-participant and move level. Source data are provided in this paper.

Code availability

All code for reproducing the statistical analyses, visualisations, training of the machine player, and execution of the experiment is available on Zenodo (<https://doi.org/10.5281/zenodo.18547122>)⁵⁹, including a static version of the experimental interface⁶⁰.

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Author contributions

L.B., T.F.E., M.D., and I.R. conceived the research project and experimental methodology; L.B., T.F.E., A.-M.N., S.B., and V.C. refined the methodology; L.B., T.F.E., A.-M.N., M.D., and I.R. developed the theoretical grounding; T.F.E. pre-registered and conducted the formal analysis; L.B. conducted the agent-based simulation; S.B. developed the machine player; V.C. implemented the experiment; L.B., T.F.E., and A.-M.N. administered and supervised the study; L.B., T.F.E., and A.-M.N. created the visualisations; L.B., T.F.E., A.-M.N., M.D., and I.R. wrote the paper.

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Competing interests

The authors declare no competing interest.

Additional information

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